DA6701 – Assignment 3
Ex-Post ML Techniques for Portfolio Optimization – Report

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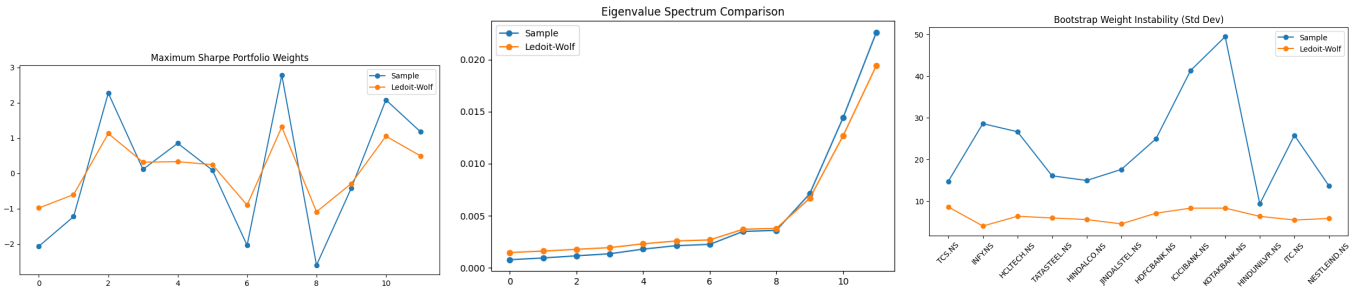
Introduction

This project uses both Ledoit-Wolf Shrinkage and Hierarchical Risk Parity to address the instability of Mean-Variance Optimization.

Ledoit-Wolf Shrinkage

- Covariance Matrix was calculated using the monthly returns, and the condition number was approximately found to be 29.34. Hence, the covariance matrix here is ill-conditioned, and small errors can amount to huge changes.
- The Ledoit–Wolf estimator improves the covariance matrix estimate by shrinking the sample covariance matrix S toward a structured target matrix F, here, a scaled identity matrix.
- The estimator takes the form $\Sigma_{LW} = (1-\delta)S + \delta F$, where $\delta \in [0,1]$ is the shrinkage intensity controlling the balance between the sample covariance and the target matrix. For our case this shrinkage intensity comes out to 0.164.
- The optimal shrinkage parameter δ is computed analytically from the data to minimize the expected squared error between the estimated covariance matrix and the true covariance matrix, reducing estimation noise especially in high-dimensional settings.
- To understand the stability of the covariance estimates, we used a bootstrap procedure in which the return data was repeatedly resampled with replacement and a covariance matrix was computed for each resample.
- This allowed us to observe the variability in covariance estimates across different samples, giving an empirical measure of the noise present in the sample covariance matrix.
- The variability estimated from these bootstrap samples helps justify the use of shrinkage, since large fluctuations indicate that the raw sample covariance is unstable and benefits from regularization toward the target matrix.



(a) (b) (c)

Figure 1: Ledoit–Wolf shrinkage and bootstrap analysis

Hierarchical Risk Parity

- The objective of this analysis is to construct and compare portfolio allocations using **Hierarchical Risk Parity (HRP)** and the traditional **Minimum Variance portfolio (MPT)**. HRP avoids direct matrix inversion and instead uses hierarchical clustering of assets based on their correlations to allocate risk more robustly.
- We first compute the covariance and correlation matrices from the training return data. Using the correlation matrix, a distance metric is constructed and hierarchical clustering is performed to identify groups of assets that behave similarly. This structure is visualized using a dendrogram and a quasi-diagonalized correlation matrix.
- Based on the clustering order, HRP assigns portfolio weights through a recursive bisection procedure that allocates capital according to the relative variance of asset clusters. For comparison, the classical minimum variance portfolio is computed using the inverse covariance matrix.
- The portfolio weights obtained from the training data are then compared with weights computed from the test period. This allows us to analyze the **stability of the allocation** by measuring how much the weights change between the two periods.
- Finally, the total portfolio turnover (sum of absolute weight changes) is evaluated for both HRP and MPT. Lower turnover indicates a more stable portfolio construction method, and the comparison highlights the robustness advantage of HRP over the traditional minimum variance approach.

Visualisation 1: Ledoit-Wolf Analysis

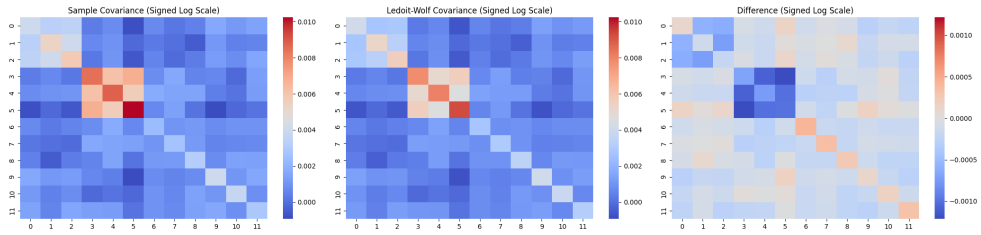


Figure 2: Heatmap of (a) Without LW, (b) With LW and (c) Change between with and Without LW

Visualisation 2: HRP Cluster Discovery

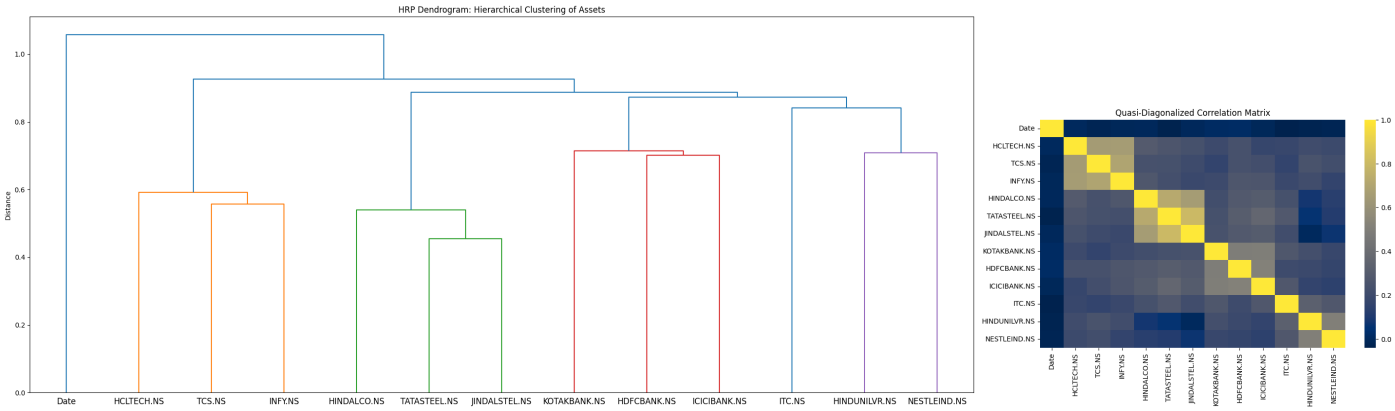


Figure 3: (a) HRP Dendogram with labelled statistical clusters, (b) Quasi-Diagonalised Covariance Matrix

Visualisation 3: Weight Sensitivity Plot

- During the March 2023 banking crisis, the MPT model (red dashed) swung erratically between 15% and 60% due to an ill-conditioned covariance matrix.
- In contrast, the HRP model (solid line) remained stable by leveraging asset topology to filter noise and minimize turnover.

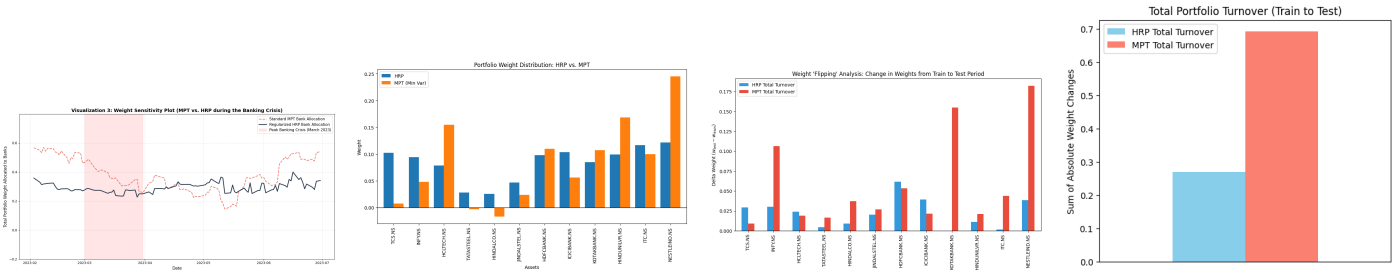


Figure 4: (a) Weight Sensitivity Plot, (b)Weight Distribution, (c)Change in weights from train to test, (d)Total Portfolio Turnover

Summary

- In a non regularized model, the high turnover would erode all alpha. HRP is stable and preserves capital.
- We don't bet on random noise here, as sample statistics are noisy and shrinkage gets rid of this issue.
- Regularization ensures our portfolio is well diversified across risk clusters than just assets, protecting us from sector-specific crisis.